

A Two-Point Counterexample to a Nonlinear Lipschitz-Summing Factorization Converse

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Status. Candidate counterexample, likely valid, subject to human review.

1 Source problem

Botelho, Maia, Pellegrino, and Santos consider an arbitrary nonempty set X , a metric space Y , a compact Hausdorff space K , an arbitrary map $\Psi : X \rightarrow C(K)$, and a map $u : X \rightarrow Y$ [1]. They call u Ψ -Lipschitz p -summing if some $C \geq 0$ satisfies

$$\left(\sum_{j=1}^m d_Y(u(x_j), u(q_j))^p \right)^{1/p} \leq C \sup_{\varphi \in K} \left(\sum_{j=1}^m |\Psi(x_j)(\varphi) - \Psi(q_j)(\varphi)|^p \right)^{1/p} \quad (1)$$

for every finite family $x_j, q_j \in X$.

Corollary 2.5 says that u is Ψ -Lipschitz p -summing if there are a probability measure μ on K and a Lipschitz map $\tilde{u} : L_p(K, \mu) \rightarrow Y$ satisfying

$$\tilde{u} \circ j_p \circ \Psi = u.$$

On PDF page 6, the authors ask whether the converse holds for $p = 2$.

$$\begin{array}{ccccc} & \sim \text{---} \text{---} \text{---} & & \sim \text{---} \text{---} \text{---} & \\ & \uparrow \Psi & & \downarrow B & \\ X & \xrightarrow{u} & Y & \xrightarrow{J} & \ell_\infty(\Gamma). \end{array}$$

The conclusion follows from Theorem 2.3 □

The converse of the corollary above holds for absolutely 2-summing *linear* operators (see [5 Corollary 2.16]). We do not know if the same holds in the nonlinear setting. It is not difficult to check that, if u is Ψ -Lipschitz 2-summing and, in addition, Y is complete and in Theorem 2.3(d), Z can be supposed to be a linear space, B to be linear and $J(Y)$ to be a subspace of Z , then there exists a Lipschitz map $\tilde{u} : L_2(K, \mu) \rightarrow Y$ such that the diagram

$$\begin{array}{ccc} C(K) & \xrightarrow{j_2} & L_2(K, \mu) \\ \uparrow \Psi & & \downarrow \tilde{u} \\ X & \xrightarrow{u} & Y \end{array}$$

commutes. But, as announced in Remark 2.4 we will prove in the next section that the map B cannot be supposed to be linear. So, we have the:

Open problem. Does the converse of Corollary 2.5 hold for $p = 2$?

2 Proof intuition

When K is a singleton, the proposed factorization space is forced to be the connected line $L_p(K) = \mathbb{R}$. The definition of Ψ -Lipschitz summability, however, permits a disconnected target. Map the two points -1 and 1 identically into the two-point metric space. The summing inequality then holds with equality, but any factorization would continuously map the connected line onto a set containing both target points. That is impossible.

3 Counterexample

Theorem 1. *For every $1 \leq p < \infty$, the converse of Corollary 2.5 in [1] is false. This remains true with K a singleton and Y a complete two-point metric space. In particular, the source open problem for $p = 2$ has a negative answer.*

Proof. Work over the real scalar field. Let $K = \{*\}$. Identify $C(K)$ with $\mathbb{R}$ by evaluation at $*$. Put

$$X = Y = \{-1, 1\}, \quad d_Y(s, t) = |s - t|,$$

and define

$$\Psi(x)(*) = x, \quad u(x) = x.$$

The target Y is finite and therefore complete.

For arbitrary $m \geq 1$ and $x_j, q_j \in X$, we have

$$\begin{aligned} \left(\sum_{j=1}^m d_Y(u(x_j), u(q_j))^p \right)^{1/p} &= \left(\sum_{j=1}^m |x_j - q_j|^p \right)^{1/p} \\ &= \sup_{\varphi \in K} \left(\sum_{j=1}^m |\Psi(x_j)(\varphi) - \Psi(q_j)(\varphi)|^p \right)^{1/p}. \end{aligned}$$

Thus u is Ψ -Lipschitz p -summing with constant one.

There is exactly one regular Borel probability measure μ on K , namely $\mu(\{*\}) = 1$. Evaluation at $*$ identifies $L_p(K, \mu)$ isometrically with $\mathbb{R}$, and under this identification $j_p \Psi(-1) = -1$ and $j_p \Psi(1) = 1$.

If the converse of Corollary 2.5 held, there would be a Lipschitz map $\tilde{u} : \mathbb{R} \rightarrow Y$ satisfying

$$\tilde{u}(-1) = -1, \quad \tilde{u}(1) = 1.$$

Every Lipschitz map is continuous. Since $\mathbb{R}$ is connected, its image under $\tilde{u}$ must be connected. But the only connected subsets of the discrete two-point space Y are singletons, whereas the displayed conditions force the image to contain both points. This contradiction proves the result. $\square$

4 Scope, verification, and novelty

The counterexample uses precisely the source hypotheses: X may be an arbitrary nonempty set and Y an arbitrary metric space. It therefore answers the printed problem, and actually works for every finite p . It does not address a strengthened Banach-valued or absolute-retract-valued version, for which the connectedness obstruction is unavailable.

The verification report rechecks the summing identity, uniqueness of the probability measure, and the topology of the forced extension. A bounded search on 9 August 2026 covered the run's cheap indexes and parsed arXiv corpus, plus exact-phrases and close web searches for the source problem. The search found the source paper and later papers citing it, but no exact answer or matching counterexample. Novelty confidence is moderate pending expert review.

References

- [1] G. Botelho, M. Maia, D. Pellegrino, and J. Santos, *A unified factorization theorem for Lipschitz summing operators*, Quarterly Journal of Mathematics **70** (2019), 1521–1533; arXiv:1902.02569.